Operators

Operators are special symbols or keywords used to perform operations on variables and values.

Tupes:

1.Arithmetic operators

2.Assignment operators

3.Comparison operators

4.Logical operators

5.Identity operators

6.Membership operators

7.Bitwise operators

1.Arithmetic operators

Arithmetic operators are used with numeric values to perform common mathematical operations.(+,-,*,/,//,%,*)

1.Addition

In [8]:
```
a=10
b=20
print(a+b)
```

```
30
```

2.subtractio

In [11]:
```
a=10
b=20
print(a-b)
```

```
-10
```

3.multiplication

In [14]:
```
a=10
b=20
print(a*b)
```

```
200
```

4.division

In [19]:
```
a=10
b=5
print(a/b)
```

```
2.0
```

5.floor division - whole number

In [22]:
```
a=10
b=5
print(a//b)
```

```
2
```

6.modules - reminder

In [25]:
```
a=10
b=5
print(a%b)
```

```
0
```

7.exponent

In [28]:
```
a=2
b=2
print(a**b)
```

```
4
```

2.Assignment Operators

Assignment operators are used to assign values to variables (=,+=,-=,=,/=,//=,%=,*=,&=,|=,^=,>>=,<<=,:=)

1.=

In [35]:
```
a=10
print(a)
```

```
10
```

2.+=

In [38]:
```
a=10
a+=20
print(a)
```

```
30
```

3.-=

In [41]:
```
a=5
a-=2
print(a)
```

```
3
```

4.*=

In [44]:
```
a=5
a*=2
print(a)
```

```
10
```

5./=

In [47]:
```
a=10
a/=5
print(a)
```

```
2.0
```

6.//=

In [50]:
```
a=10
a//=5
print(a)
```

```
2
```

7.%=

In [53]:
```
a=10
a%=5
print(a)
```

```
0
```

8.&=

In [56]:
```
a=5
a&=3
print(a)
```

```
1
```

9.|=

In [59]:
```
a=5
a|=3
print(a)
```

```
7
```

10.^=

In [62]:
```
a=5
a^=3
print(a)
```

```
6
```

11.>>=

In [65]:
```
a=3
a>>=2
print(a)
```

```
0
```

12.<<=

In [68]:
```
a=3
a<<=2
print(a)
```

```
12
```

13.**=

In [71]:
```
x = 5
x **= 3
print(x)
```

```
125
```

3.Comparision

Comparison operators are used to compare two values (==,!=,>,<,>=,<=)

1. equal(==)

In [76]:
```
a=3
b=3
print(a==b)
```

```
True
```

2.not equal(!=)

In [79]:
```
a=4
b=9
print(a!=b)
```

```
True
```

3.greater than(>)

In [82]:
```
a=4
b=9
print(a>b)
```

```
False
```

4.less than

In [85]:
```
a=4
b=9
```

```
print(a<b)
```

```
True
```

5.greater than or equal to(>=)

In [88]:
```
a=4
b=9
print(a>=b)
```

```
False
```

6.less that or equal to(<=)

In [92]:
```
a=4
b=4
print(a<=b)
```

```
True
```

4.logical operator

Logical operators are used to combine conditional statements(and , or , not)

1.and - return true when both condition is True

In [97]:
```
a=10
print(a>5 and a<20)
```

```
True
```

2.or - return true atleast one condition is True

In [100…
```
a=10
print(a<5 or a<20)
```

```
True
```

3.not - opposite of and

In [103…
```
a=10
print(not(a>5 and a<20))
```

```
False
```

5.Bitwise operator

Bitwise operators are used to compare (binary) numbers (&,|,^,>>,<<,~)

1.&

In [108…
```
a=5
b=3
print(a&b)
```

```
1
```

2.|

In [111…
```
a=5
b=3
print(a|b)
```

```
7
```

3.^

In [114…
```
a=5
b=3
print(a^b)
```

```
6
```

4.>> - right shift

In [117…
```
a=5
b=3
print(a>>b)
```

```
0
```

5. << - left shift

In [121…
```
a=5
b=3
print(a<<b)
```

```
40
```

6.~

In [124…
```
a=~5
print(a)
```

```
-6
```

6.Membership operators

Membership operators are used to test if a sequence is presented in an object.(in , not in)

1.in

In [129…
```
a=("Hello","Hi","bye")
print("Hi"in a)
```

```
True
```

In [131…
```
a="ryt kji iwd"
print("kji" in a)
```

```
True
```

2.not in

In [134…
```
a=("Hello","Hi","bye")
print("Eyee" not in a)
```

```
True
```

7.Identity operators

Identity operators are used to compare the objects, not if they are equal, but if they are actually the same object, with the same memory location.(is , is not)

1.is

In [141…
```
a=[1,2,3]
b=a
print(a is b)
```

```
True
```

In [145…
```
# it allow 256 memory
a=20
b=20
print(a is b)
```

```
True
```

In [152…
```
a=10
b=20
print(a is b)
```

```
False
```

2.is not

In [148…
```
a=[1,2,3]
b=[1,2,3]
print(a is not b)
```

```
True
```

8.Ternary Operator

The ternary operator allows you to assign one value if a condition is true, and another if it is false

In [156…
```
a=100
b=20
print("a is larger"if a>b else "b is larger")
```

```
a is larger
```

Practice Questios

1. Write a Python program to perform addition, subtraction, multiplication, and division on two numbers.

In [160…
```
num1 = int(input("enter num1:"))
num2 = int(input("enter num2:"))
add = a+b
print("addition:",add)
sub = a-b
print("subtraction:",sub)
multi = a*b
print("multiplication:",multi)
div = a/b
print("division:",div)
```

```
addition: 120
subtraction: 80
multiplication: 2000
division: 5.0
```

2. Write a Python program to find the remainder when one number is divided by another using the % operator.

In [165…
```
num1 = 10
num2 = 3
print(num1 % num2)
```

```
1
```

3. Write a Python program to find the square of a number using the ** operator.

In [168…
```
a = 5
b= 2
print( a**b)
```

```
25
```

4. Write a Python program to swap two variables using arithmetic operators only.

In [175…
```
a = 10
b = 20
a = a+b
b = a-b
a=a-b
print(a)
print(b)
```

```
20
10
```

5. Write a Python program to check the output of 10 + 5 * 2.

In [178…
```
a=10
b= 5*2
print(a+b)
```

```
20
```

6. Write a Python program to compare two numbers using comparison operators (==, !=, >, <).

In [181…
```
a=23
b=55
print(a==b)
print(a!=b)
print(a>b)
print(a<b)
```

```
False
True
False
True
```

7. Write a Python program to check the result of logical operators (and, or, not) using Boolean values.

In [186…
```
a=True
b=False
print(a and b)
print(a or b)
print(not a)
print(not b)
```

```
False
True
False
True
```

8. Write a Python program to perform bitwise AND (&), OR (|), and XOR (^) on two integers.

In [189…
```
a=3
b=2
print(a & b)
print(a | b)
print(a ^ b)
```

```
2
3
1
```

9. Write a Python program to calculate the quotient and remainder of two numbers using /, //, and %.

In [192…
```
a=5
b=10
print(a/b)
print(a//b)
print(a%b)
```

```
0.5
0
5
```

10. Write a Python program to increase a variable's value by 5 using an assignment operator (+=).

In [195…
```
a=10
a += 5
print(a)
```

```
15
```